C2SM Git for Advanced Workshop 2025

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Center for Climate
Systems Modeling

12 September 2025

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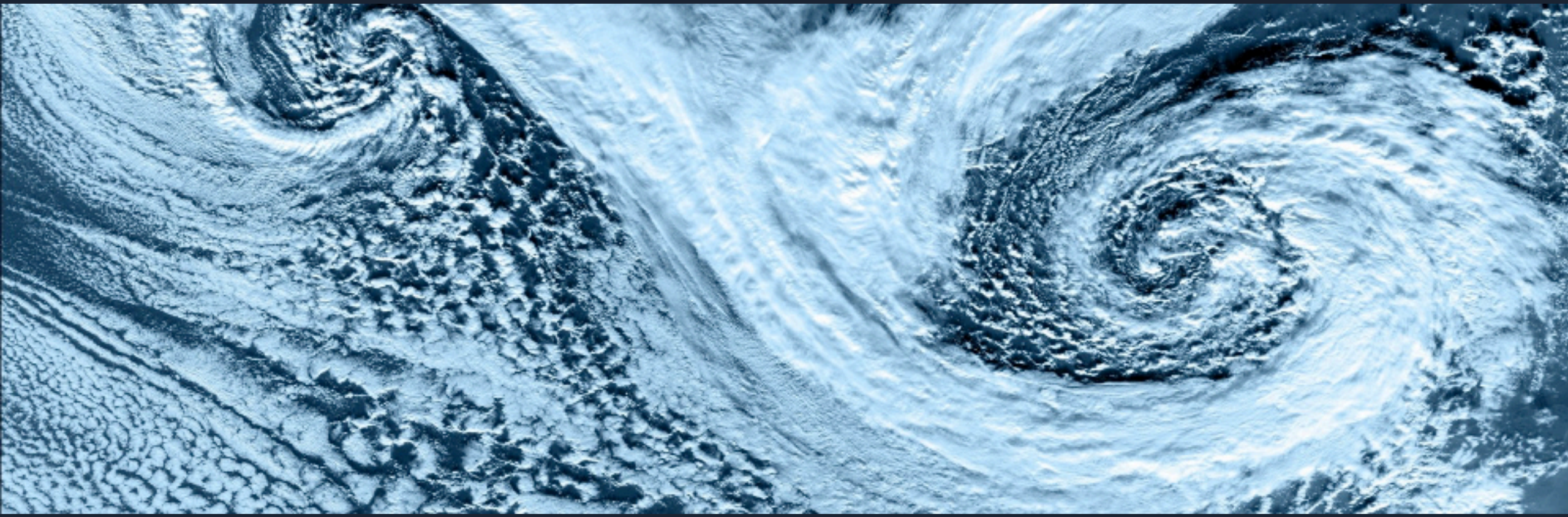

Outline

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Part 0: Recap on Git

- Why use Git?
- Practical example
- Local Git workflow

Part 1: Examining a Git Repository

- Useful commands to examine Git repositories
- Exercises 1-2

Part 2: Git Workflow

- Web interface workflow
- Web interface demonstration
- Useful workflow commands
- Git cherry-pick
- Custom Git Hooks
- Exercises 3-7

Part 3: Nesting Git Repositories

- Using Git submodules
- Exercise 8

Part 4: Useful Tools and Resources

- External Git tools
- Git resources
- Git in VS Code demonstration

Schedule

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09:00 – 09:10 Welcome & Git Recap

09:10 – 10:00 Exercise 1 – 2

10:00 – 11:00 Exercise 3 – 5

11:00 – 11:20 Coffee Break

11:20 – 11:40 Exercise 6 – 7

11:40 – 12:10 Exercise 8

12:10 – 12:30 Useful Tools demonstration

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Part 0:

Git Recap

Why Use Git?

- Tracks file changes in a documented way

Without versioning

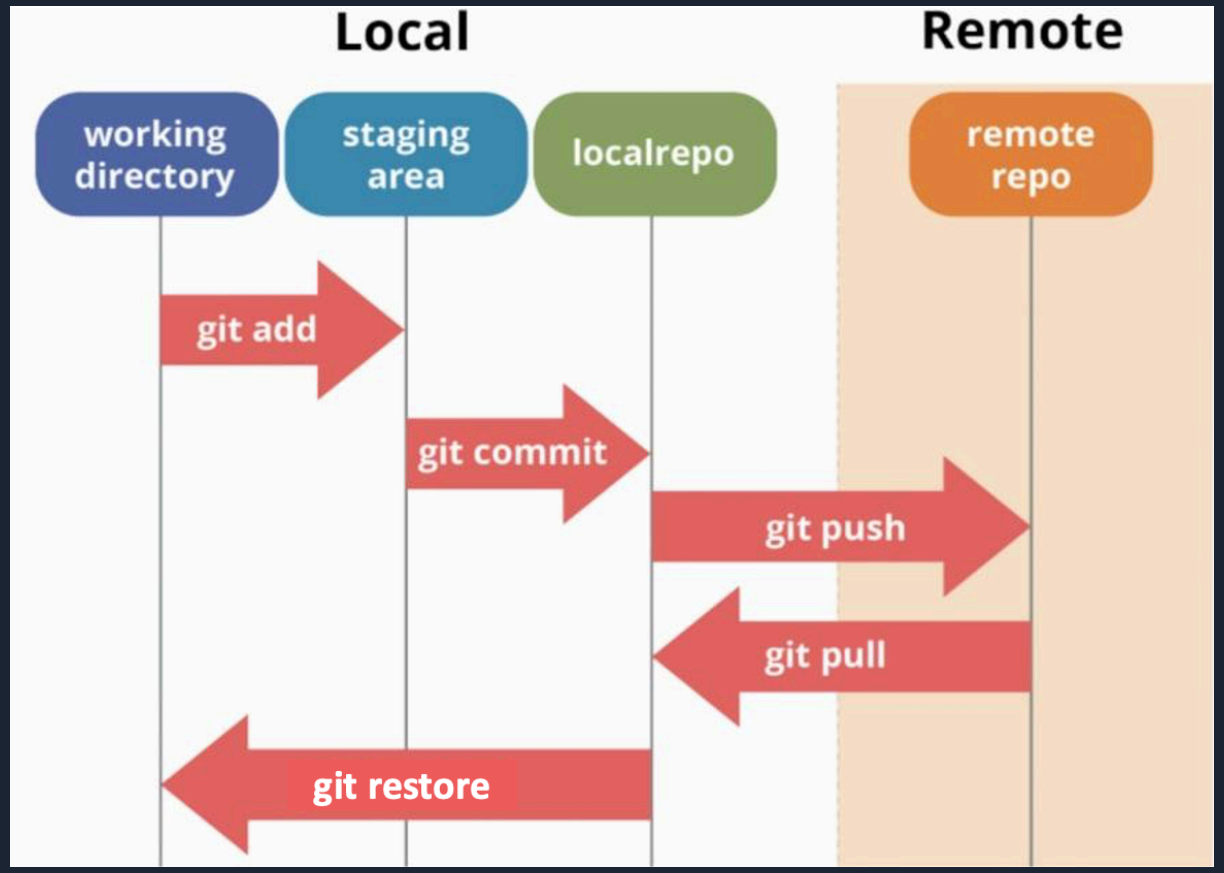
- conference_schedule_v0.txt
- conference_schedule_v1.txt
- conference_schedule_v2.txt
- conference_schedule_v2_dj.txt
- conference_schedule_v2_orch.txt
- conference_schedule_v3.txt

With versioning

- conference_schedule.txt
- .git (file, which contains information of all changes including reasons for changes, if well documented)

- Allows us to work simultaneously on the same code (when using remote server)
  - Alone (on different computers)
  - Multiple people in a collaboration
- Maintain several parallel versions of the same code in a systematic way.
- Many tools available (web-based services, graphical interfaces, etc.)

Local Workflow Recap



Source: https://dev.to/mollynem/git-github--workflow-fundamentals-5496

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Part 1:

Examining a Git Repository

Useful Commands

`git log`

shows the commits in a repository

`git blame`

shows when what part of file was changed last by which commit

`git diff`

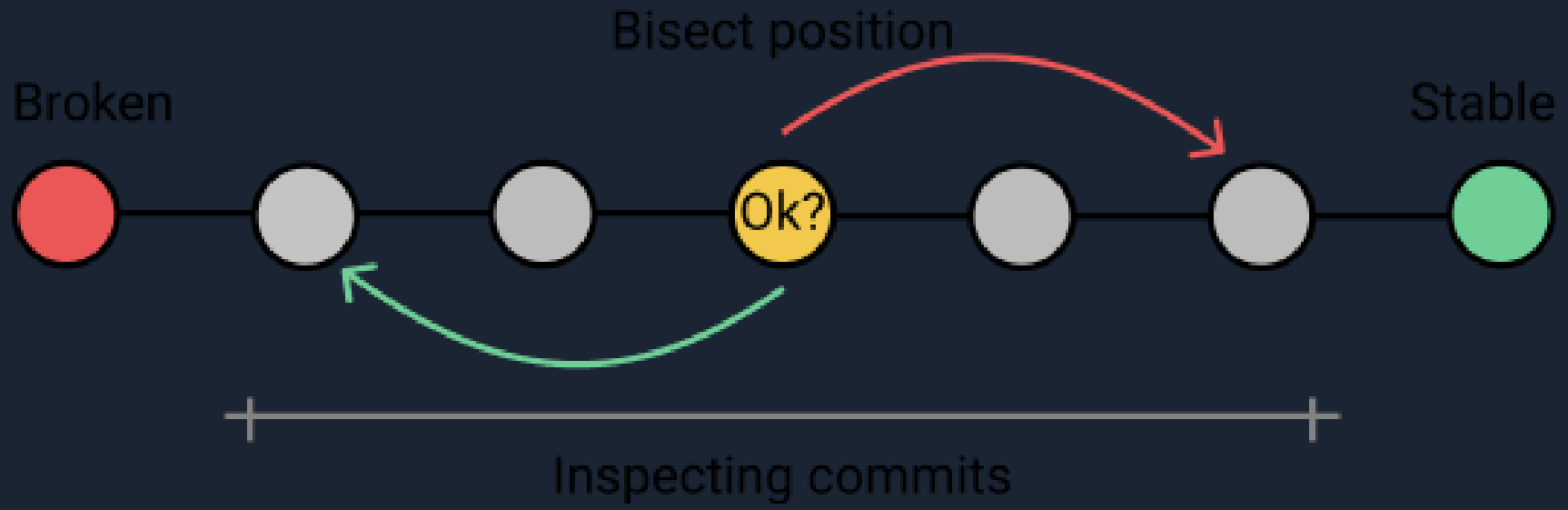
shows changes between commits, commit and working tree, etc.

`git show`

shows both commit information AND commit diff

These commands have many different options for customizing the output (explored in Exercise 1)

git bisect



- Iteratively locate the commit where a change occurred
- Requires a linear history to work correctly

Examining a Git Repository: Exercises

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- Exercise 1: `git log`, `git blame`, `git diff`, and `git show`
- Exercise 2: `git bisect`

Exercises can be found at: https://github.com/C2SM/git-course/tree/main/advanced

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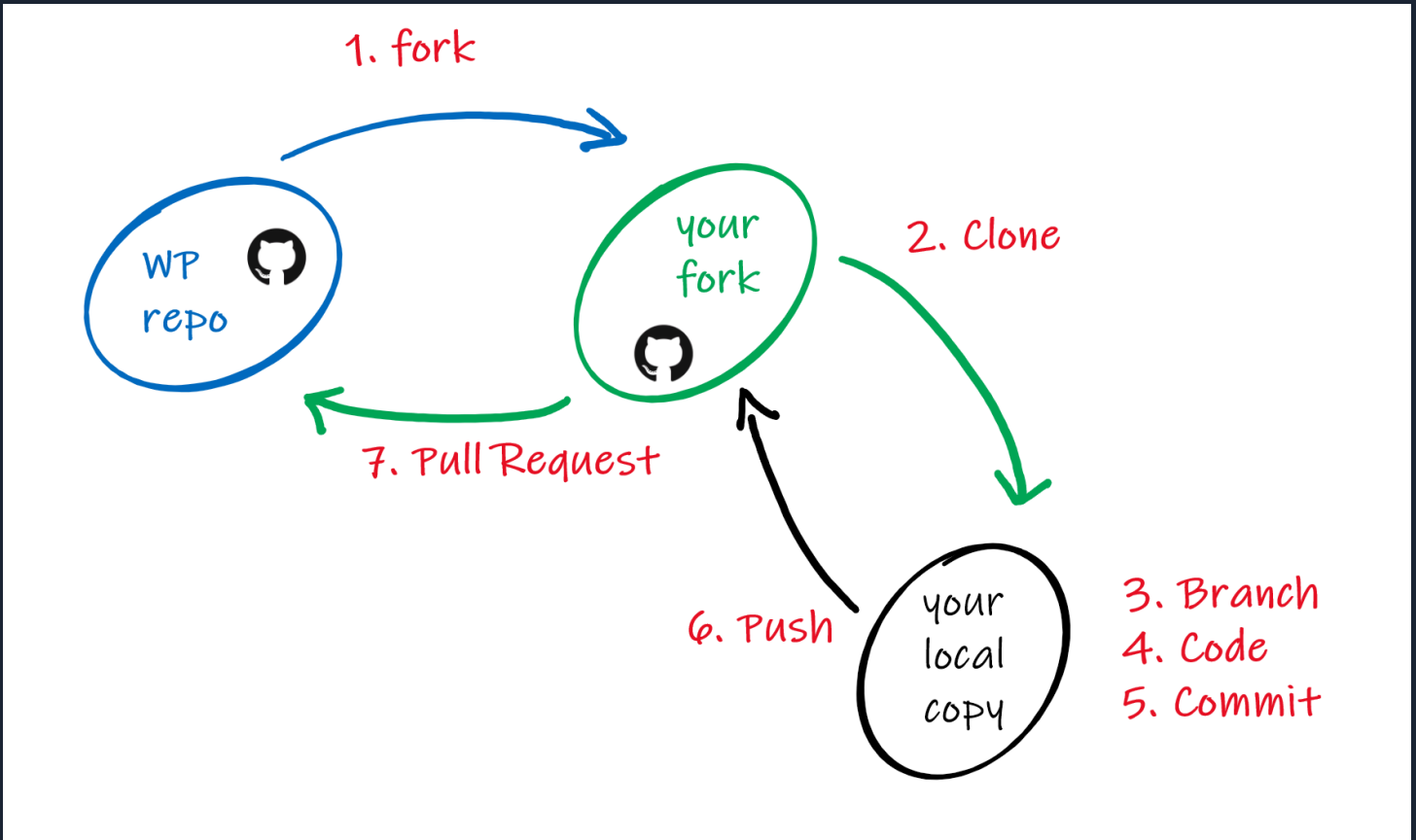
Part 2:

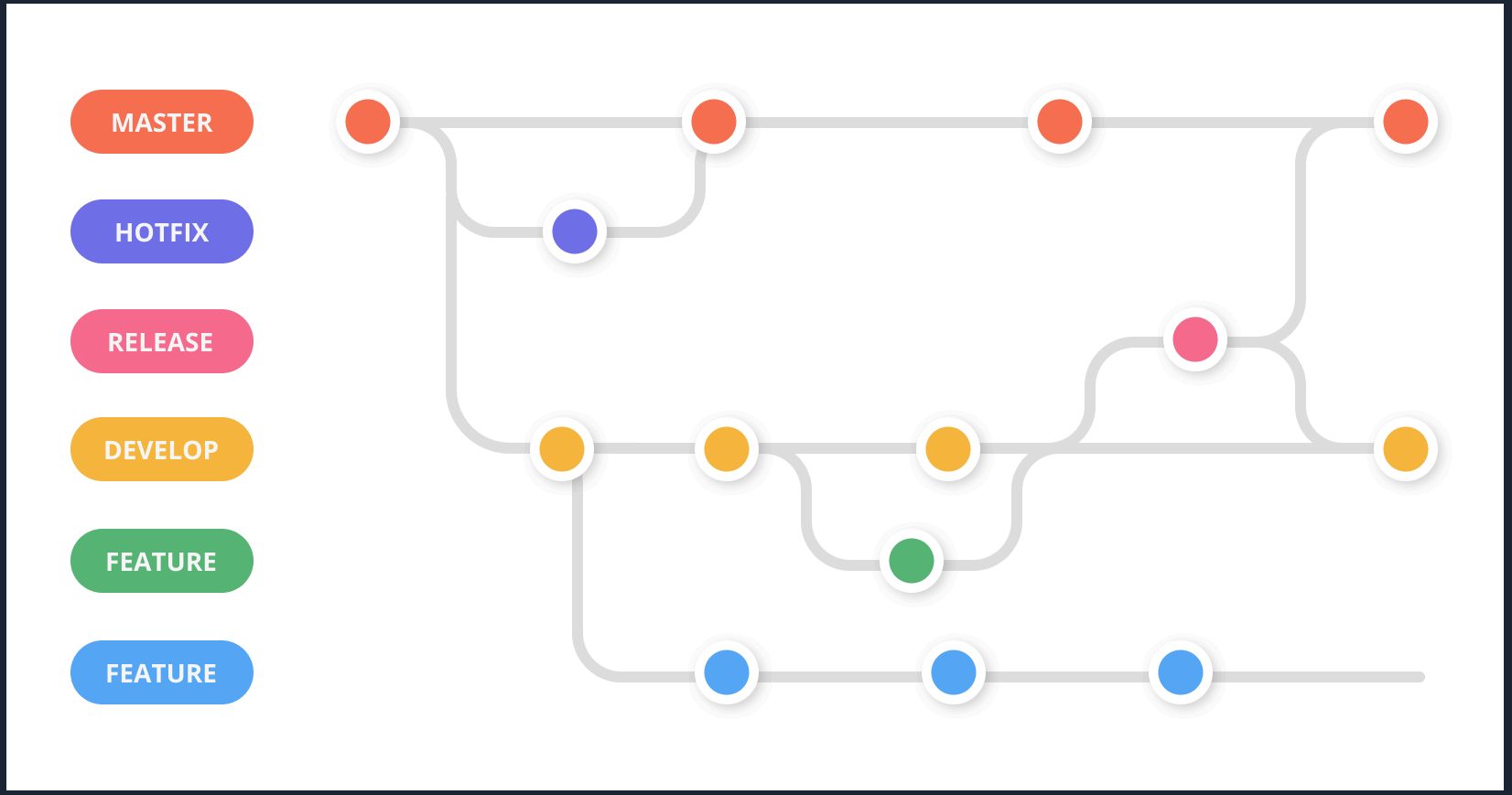
Git Workflow

Web Interface Workflow – Repository Level



Source: https://developer.wordpress.org/block-editor/contributors/code/git-workflow/

Web Interface Workflow - Branch Level



.gitignore

- Tell Git to disregard files you don't want committed.
- Best practice is to ignore binaries, intermediate files, files that can be generated from files in your repository, etc.

```
*~
*.exe
netcdf-*
bin
!bin/gen_info.sh
```

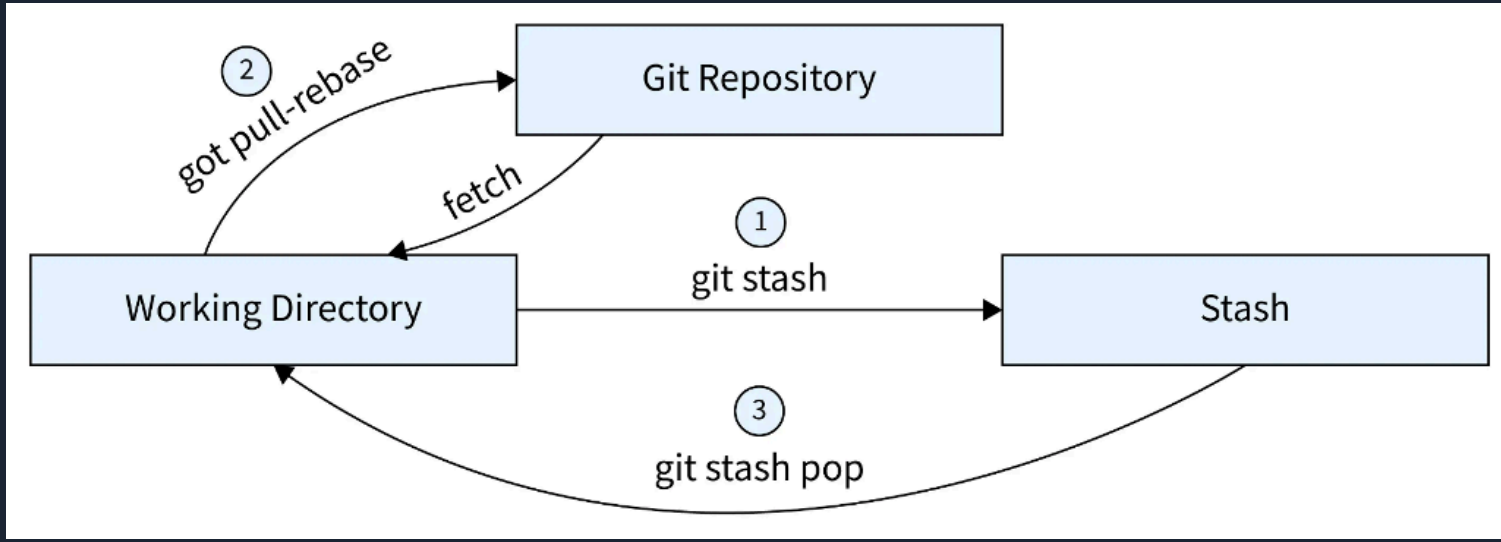
.gitkeep

- Git keeps track of files, not folders
- Put an empty `.gitkeep` file in any folder you would like to keep in the repository
- Commit the `.gitkeep` file to the Git repository

Note: This is a convention that has developed, not an official Git feature like `.gitignore`.

git stash

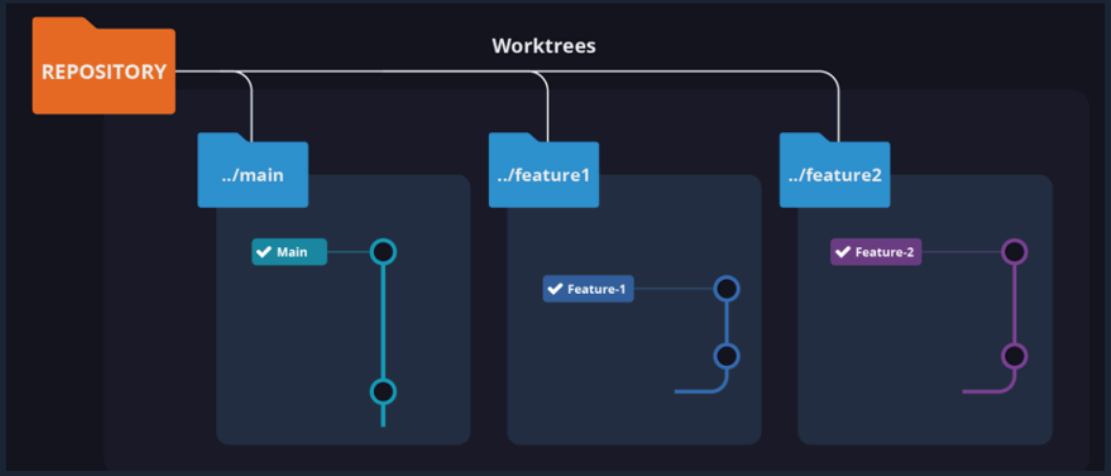
- Allows you to save bits of work without committing them and reuse them late
- Useful when:
  - you need to pull changes, but have uncommitted changes
  - you need to switch branch, but have uncommitted changes



Source: https://www.scaler.com/topics/git/git-stash-pop/

git worktree

- Can checkout and work with multiple branches of a repository with a single clone
- Worktrees share a single `.git` directory, which:
  - **saves memory and time** compared to multiple clones
  - keeps the git configuration **centralized**
- Can build/test multiple branches simultaneously



Source: https://www.gitkraken.com/learn/git/git-worktree

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GitHub Web Interface Demonstration

https://github.com/C2SM/git-workflow-practice

Git Workflow - Exercises

- Exercise 3: `.gitignore`
- Exercise 4: `git stash` and `git worktree`
- Exercise 5: practice the git workflow

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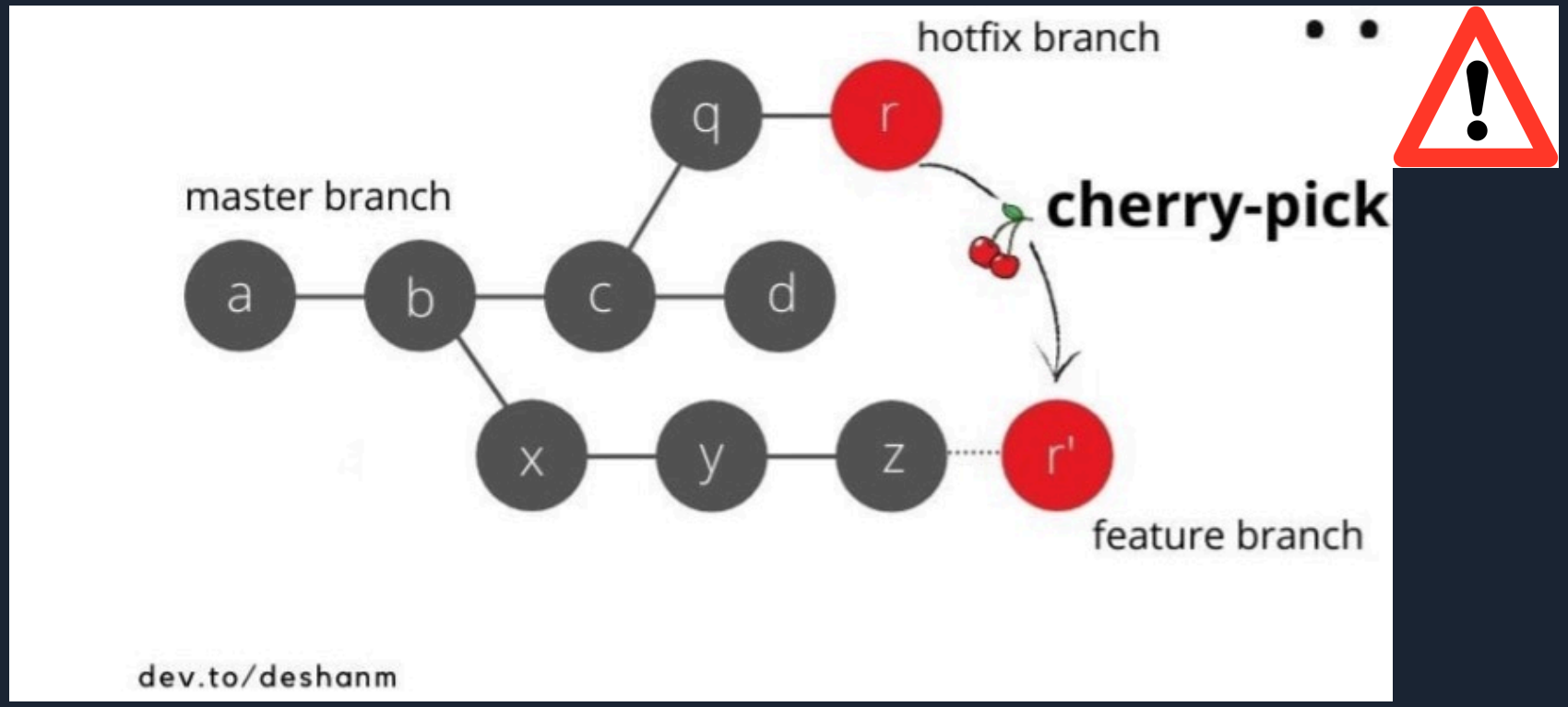
Coffee Break

git cherry-pick: Snagging one Commit



- Grabs one commit and puts it at the head of another branch.
- Uses a different commit ID for the same file changes.

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Custom Git Hooks

Custom Git Hooks

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- Scripts to automate / enforce certain actions
- Triggered when certain (pre-defined) events occur
- Stored in `.git/hooks`
- Named after the event they are associated with (e.g., `pre-commit`, `post-merge`, etc.)
- Can be used for
  - enforcing coding standards
  - preventing accidental commits of sensitive data
  - triggering automatic tests
  - updating documentation
- Samples already present!

Git Workflow - Exercises

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- Exercise 6: `git cherry-pick`
- Exercise 7: Custom Git Hooks

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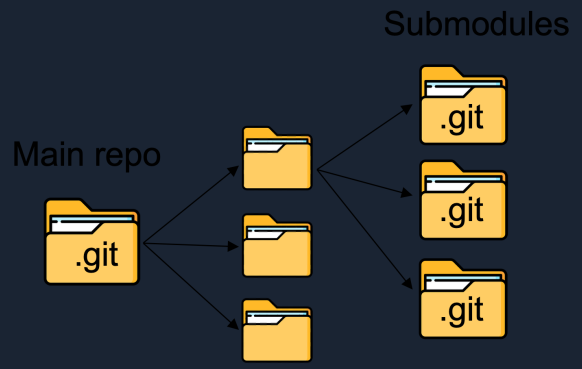
Part 3:

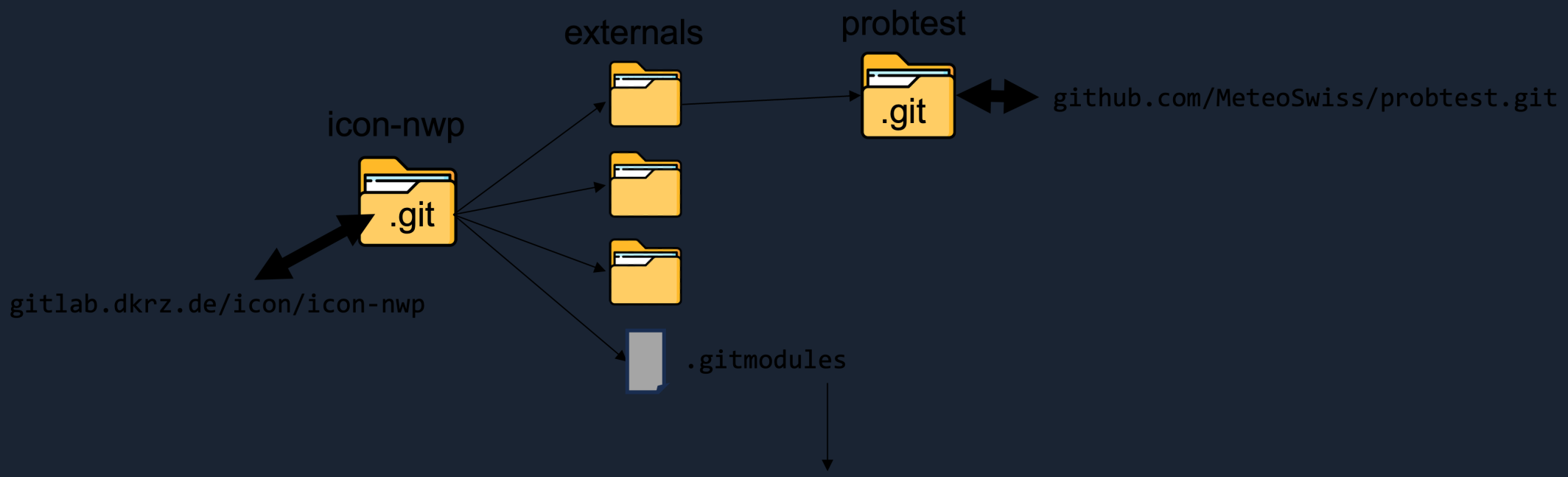
Nesting Git Repositories

Nested Repositories



- Why use it?
  - **Modularity**: Break project into smaller, manageable pieces.
  - **Version Control**: Each submodule has its own Git history and version tracking
  - **Collaboration**: Multiple teams can work on submodules independently
- Options:
  - **Git Submodules**: Git's built-in mechanism
  - **Git Subtrees**: Alternative approach

Parent repository stores reference to external repositories



Cloning a Repository with Submodules

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`git clone`

By default, Git does NOT clone contents of submodules

`git clone --recurse-submodules`

Check out contents of any submodules when cloning parent repo

`git submodule update --init`

Get contents of submodules after cloning

Git Submodules

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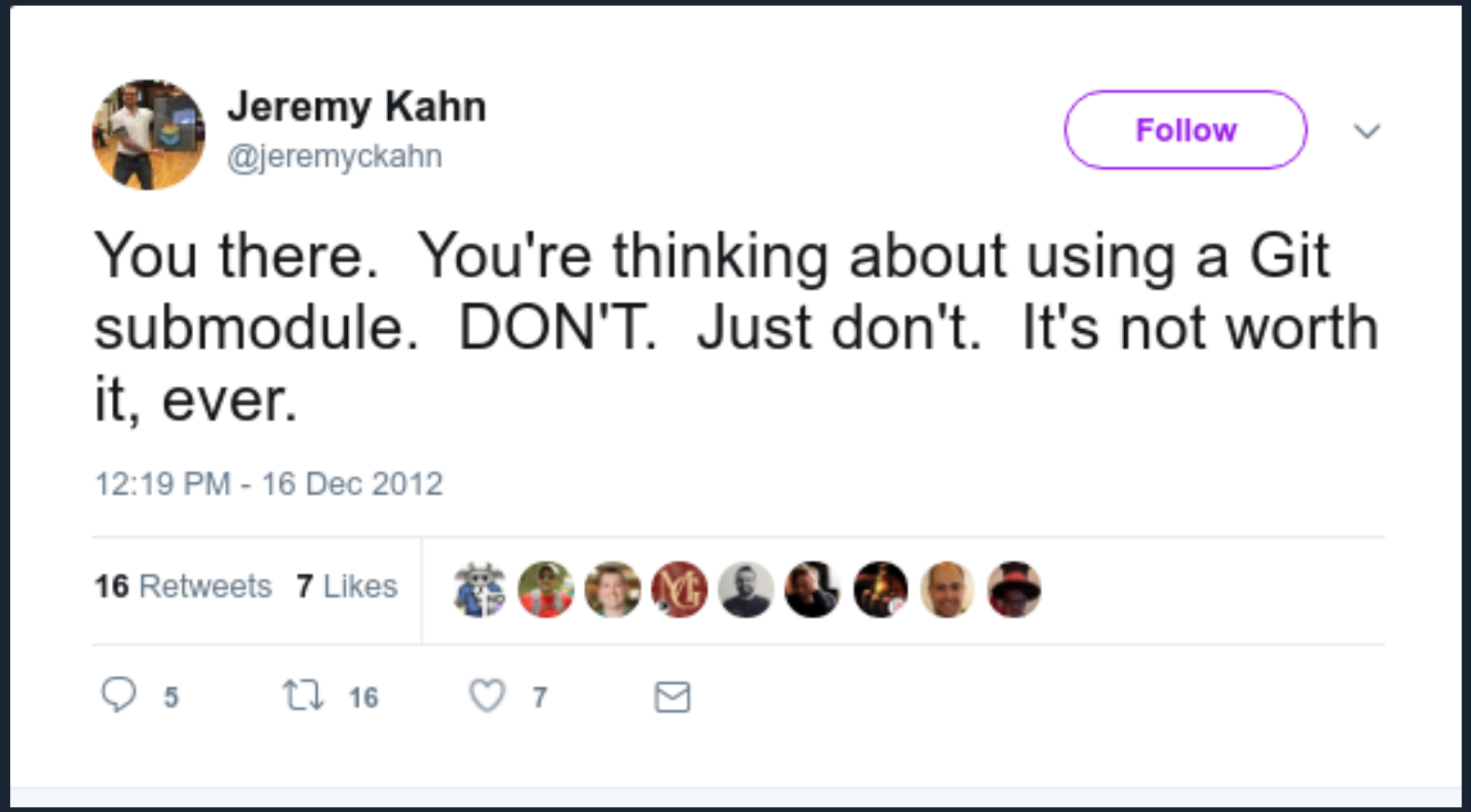

Nesting Git Repositories — Exercises

- Exercise 8: `git submodule`

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Part 4:

Useful Tools and Resources

Wide Range of Available Tools

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Text editor plugins:

- magit (Emacs): Wrapper for git commands
- vim-gitgutter (vim): Improved git diff viewing

Integrated development environment (IDE) integration:

- Visual Studio Code
- RStudio
- Eclipse

Official Git tools:

- git-gui: Focuses on commit generation
- gitk: Focuses on displaying diffs

Terminal prompt changer

- fancy-git

Git GUIs:

- Git for Windows: Git BASH command line
- TortoiseGit: Windows Shell Interface to Git

Git Resources

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- Official Git manual: http://git-scm.com/
- GitHub manual: https://docs.github.com
- Cheat sheet with useful GitHub commands for quick reference: https://education.github.com/git-cheat-sheet-education.pdf

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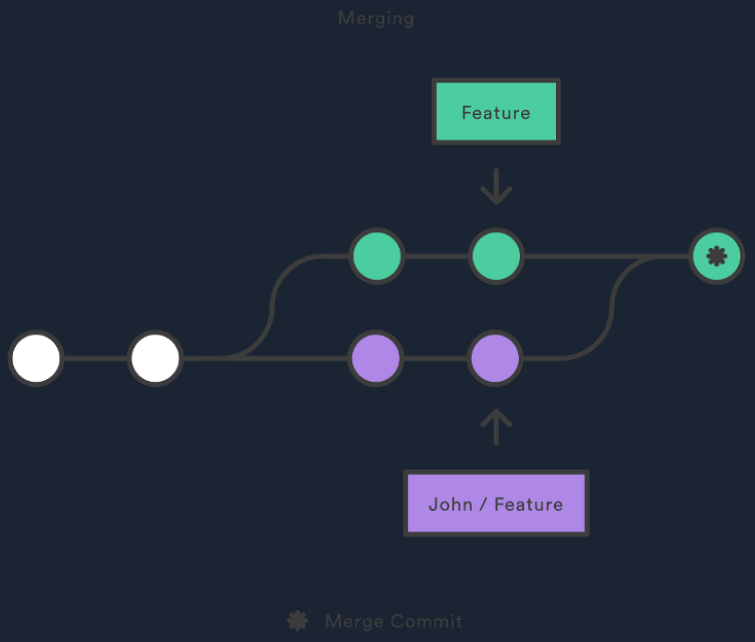
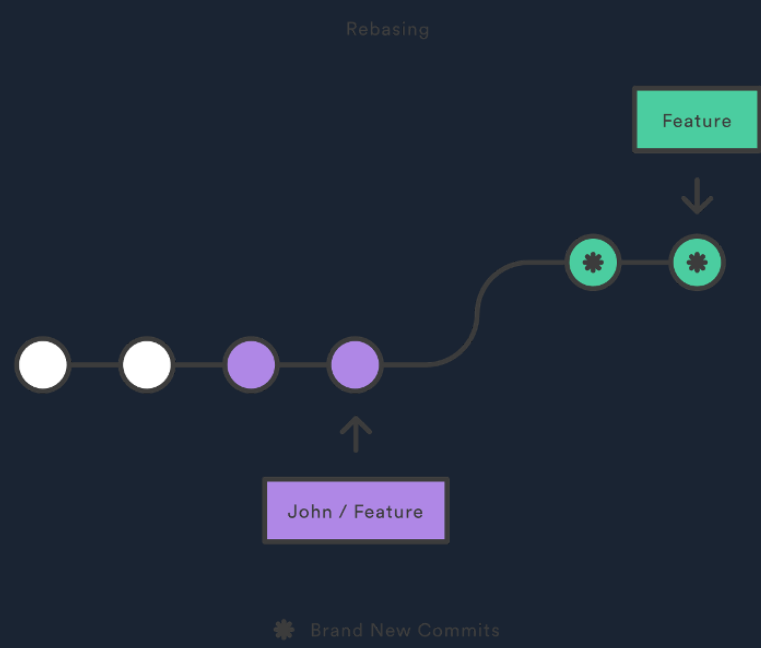
Bonus Part

git rebase: Alternative to git merge





Source: https://dzone.com/articles/merging-vs-rebasing

- Commits are replayed in a different order
- Advantage: keep cleaner commit history
- Should NEVER be used in a shared branch

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Questions? Comments?